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(ii) Suppose

SI contains 100h

DI contains 200h

AX contains 4142h

Direction Flag= 0

Byte 100h contains 10h

Byte 101h contains 15h

Byte 200h contains 20h

Byte 201h contains 25h

Give the value moved if MOVSB instruction is executed. Also give the new contents of SI and DI.
[3]

Value Moved: **10h**

SI: **101h**

DI: **201h**

(iii) Given the code below, answer the following questions: (The printAsterisk subroutine takes x-coordinate, y coordinate, attribute and character as parameters in this order).

(a) What are the background and foreground color of the asterisks printed on the screen? [1]

Background Color: **White**

Foreground Color: **Red**

(b) Give the positions (x, y) coordinates of the screen where asterisks are printed? [5]

(7,8)

(7,9)

(7,10)

(7,11)

(8,11)

(9,11)

(10,11)

(8,9)

(9,10)

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[org 0x0100]	Mov bx, [bp+4]	Disp1: mov ax, [bp+10]	Mov cx, [bp-2]	Mov cx, [bp-2]	Mov ax, [bp+12]	Start: Mov al, 0x74	Mov ax, 11; y2
Jmp start			Mov dx, -1				Push ax
TrianglePrint:	Cmp bx, [bp+8]	Mov bx, [bp+8]	Disp2: mov ax, [bp+6]	Sub cx, 1	Push ax	Push ax	Call TrianglePrint
push bp	Jbe end	Inc dx		Mov dx, 0	Call PrintAsterisk	Mov al, '*'	
Mov bp,sp		Add bx, dx	Mov bx, [bp+4]	Disp3: mov ax, [bp+10]	Loop Disp3	Push ax	Mov ax, 0x4c00
Sub sp,2	Sub bx, [bp+8]	Push ax	Inc dx		End:pop dx	Mov ax, 7; x1	Int 0x21
Push ax	Cmp ax, bx	Push bx	Sub ax, dx	Mov bx, [bp+8]	Pop cx	Push ax	
Push bx	Jne end	Mov ax, [bp+14]	Push ax	Inc dx	Pop bx	Mov ax, 8; y1	
Push cx	Mov [bp-2], ax	Push ax	Push bx	Add ax, dx	Pop ax	Push ax	
Push dx	Add ax, 1;	Mov ax, [bp+12]	Mov ax, [bp+14]	Add bx, dx	Mov sp, bp	Mov ax, 10; x2	
Mov ax, [bp+6]	Mov cx, ax	Push ax	Push ax	Push ax	Pop bp	Push ax	
Cmp ax, [bp+10]	Mov dx, -1	Call PrintAsterisk	Mov ax, [bp+12]	Push bx	Ret 12		
Jbe end		Loop Disp1	Push ax	Mov ax, [bp+14]			
Sub ax, [bp+10]			Call PrintAsterisk	Push ax			
			Loop Disp2				

CLO # 3: Apply the knowledge of Intel x86 architecture to develop moderately complex and well-modularized assembly programs.

Q2: Programming Question [15 marks]

Write a subroutine makeBorders that

1. Changes the background color of the screen to white.
2. Displays all the characters on the main border of the screen in red color.
3. Displays all the characters on the inner border in blue color with blinking.

You do not need to change the characters on the screen. The characters on the screen should remain as it is. Just change the attributes. The main and inner border is shown in the screen below.

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```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
AB                                                                    B A
BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB A
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

The positions where A is displayed is the main border of the screen and the positions where B is displayed is the inner border of the screen.



